

Date: 2019

Q-No. 1 Same 2017

Q-No. 2 order statistics

$$Y_1 < Y_2 < Y_3 < Y_4$$

$$f(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the distribution of Y_3 .

Solution:-

$$f(x) = 2x$$

$$F(x) = \int_0^x 2x \cdot dx$$

$$= 2 \left[\frac{x^2}{2} \right]_0^x$$

$$F(x) = x^2$$

The order statistics is

$$g(y_r) = \frac{n!}{(r-1)!(n-r)!} \left[F(y_r)^{r-1} f(y_r) (1-F(y_r))^{n-r} \right]$$

put $r=3$ $n=4$

$$= \frac{4!}{2!(1)!} F(y_3)^2 f(y_3) (1-F(y_3))$$

$$= \frac{24}{2 \times 1 \times 1} x (y_3^2)^2 (2y_3) (1-y_3^2)$$

$$= \frac{24}{2(12)} y_3^5 (1-y_3)$$

$$g(y_r) = 24 y_3^5 (1-y_3)$$

$$24(y_3^5 - y_3^6) \quad \star$$

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Q-NO. III let X and Y are independent random variables with joint p.d.f

$$f(x, y) = x + y \quad 0 < x < 1, 0 < y < 1$$
$$= 0 \quad \text{elsewhere}$$

Find $E(XY)$ and $E(X+Y)$

Solution:-

$$E(XY) = \dots$$

$$E(XY) = \int_0^1 \int_0^1 x, y (x+y) dx dy$$

$$\int_0^1 y \left\{ \int_0^1 x^2 dx + \int_0^1 xy dx \right\} dy$$

$$\int_0^1 \left\{ y \left[\frac{x^3}{3} \right]_0^1 + y \left[\frac{x^2}{2} \right]_0^1 \right\} dy$$

$$\int_0^1 y \frac{dy}{3} + \int_0^1 y \frac{dy}{2}$$

$$= \left[\frac{y^2}{6} \right]_0^1 + \left[\frac{y^2}{4} \right]_0^1$$

$$= \frac{1}{6} + \frac{1}{4}$$

$$\frac{2}{12} + \frac{3}{12} = \frac{5}{12}$$

$$E(XY) = f_X(y) = \int_0^1 y(x+y) dx$$

$$\int_0^1 x + y dx$$

$$\frac{x^2}{2} +$$

$$\int_0^1 x dx + \int_0^1 y dx$$

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$$\left| x^2/2 + x \right|_0^1$$

$$= 1/2 + 1$$

$$f(x)(x) = \int_0^1 x+y \, dy$$
$$= \left| xy + y^2/2 \right|_0^1$$

$$= x + 1/2$$

$$E(X) = \int_0^1 x f(x) \, dx$$

$$= \int_0^1 x \left(x + \frac{1}{2} \right) dx$$

$$= \int_0^1 x^2 + \frac{x}{2} \, dx$$

$$= \left| x^3/3 + x^2/4 \right|_0^1 \rightarrow 1/3 + 1/4$$

$$7/12$$

$$E(Y) = \int_0^1 y \left(y + \frac{1}{2} \right) dy$$

$$= \int_0^1 y^2 + y/2 \, dy$$

$$= \left| y^3/3 + y^2/4 \right|_0^1 \rightarrow 1/3 + 1/4$$

$$7/12$$

$$E(X+Y) = E(X) + E(Y)$$

$$= 7/12 + 7/12 \rightarrow 7/6$$

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Q-No. x Pre

Pareto Distribution :-

mean AND variance:

Solution:-

$$\mu_1' = E(x) = \int_k^{\infty} x \cdot f(x) dx$$

$$= \int_k^{\infty} x \cdot \frac{\alpha k^{\alpha}}{x^{\alpha+1}} dx$$

$$= \alpha k^{\alpha} \int_k^{\infty} x \cdot x^{-\alpha-1} dx$$

$$= \alpha k^{\alpha} \int_k^{\infty} x^{-\alpha} dx$$

$$= \alpha k^{\alpha} \int_k^{\infty} x^{-\alpha} dx$$

$$= \alpha k^{\alpha} \left[\frac{x^{-\alpha}}{-\alpha} \right]_k^{\infty}$$

$$= \alpha k^{\alpha} \left[\frac{k^{-\alpha}}{-\alpha} - \frac{\infty^{-\alpha}}{-\alpha} \right]$$
$$= k^{\alpha-\alpha} - \frac{\infty^{\alpha}}{\alpha} = 1$$

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Q-NU-3

lognormal Distribution and
 y th moment about origin:

Solution:-

$$\mu_y' = E(x^y) = \int_0^{\infty} x^y f(x) dx$$
$$= x^y \cdot \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \left(\log \left(\frac{x-\mu}{\sigma} \right) \right)^2} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_0^{\infty} x^y x^{-1} \cdot e^{-\frac{1}{2} \left(\log \left(\frac{x-\mu}{\sigma} \right) \right)^2} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_0^{\infty} x^{y-1} e^{-\frac{1}{2} \left(\log \left(\frac{x-\mu}{\sigma} \right) \right)^2} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_0^{\infty} x^{y-1} e^{-\frac{1}{2} \left(\log \left(\frac{x-\mu}{\sigma} \right) \right)^2} dx$$

$$\text{let } \log \left(\frac{x-\mu}{\sigma} \right) = y$$

$$\log(x-\mu) = \sigma \sqrt{2\pi} y$$

$$x = e^{\mu + \sqrt{2\pi} \sigma y}$$

$$dx = e^{\mu + \sqrt{2\pi} \sigma y} \cdot \sqrt{2\pi} \sigma dy$$

$$\text{Now } \frac{1}{\sigma \sqrt{2\pi}} \int_0^{\infty} (e^{\mu + \sqrt{2\pi} \sigma y})^{y-1} e^{-y^2} e^{\mu + \sqrt{2\pi} \sigma y} dy \cdot \sigma \sqrt{2\pi}$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{\mu(y-1) + (y-1)\sqrt{2\pi} \sigma y - y^2 + \mu + \sqrt{2\pi} \sigma y} dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{\mu + \mu + y\sqrt{2\pi} \sigma y - \sqrt{2\pi} \sigma y - y^2 - \sqrt{2\pi} \sigma y + \mu} dy$$

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$$= \frac{e^{-uy - u^2/4}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y^2 - 2y\delta/\sqrt{2} + (\frac{\gamma\delta^2}{12})^2 - (\frac{\gamma\delta}{12})^2)} dy$$

$$= \frac{e^{-uy}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y^2 + 2y\delta/\sqrt{2} + (\frac{\gamma\delta^2}{12})^2 - (\frac{\gamma\delta}{12})^2} dy$$

$$= \frac{e^{-uy}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y - \gamma\delta/\sqrt{2})^2} e^{\gamma^2\delta^2/2} dy$$

$$= \frac{e^{-uy + \gamma^2\delta^2/2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y - \gamma\delta/\sqrt{2})^2} dy$$

put $(y - \gamma\delta/\sqrt{2})^2 = z$

$$\sqrt{z} = y - \gamma\delta/\sqrt{2}$$

$$dy = \frac{1}{2\sqrt{z}} dz$$

even function

$$dy = \frac{1}{2\sqrt{z}} dz$$

$$\frac{e^{-uy + \gamma^2\delta^2/2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z} \frac{1}{2\sqrt{z}} dz$$

$$\frac{e^{-uy + \gamma^2\delta^2/2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z} z^{-1/2-1+1} dz$$

$$\frac{e^{-uy + \gamma^2\delta^2/2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z} z^{1/2-1} dz$$

$$\frac{e^{-uy + \gamma^2\delta^2/2}}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2}$$

$$r^{th} = e^{-uy + \gamma^2\delta^2/2}$$

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If $\log x$ is N.D with mean = 1 variance = 4

FIND $P(1/2 < x < 2)$

$$\mu_1 = 1 \quad \sigma^2 = 4 \quad \sigma = 2$$

$$f(x) = \frac{1}{x\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\log \frac{x-\mu}{\sigma}\right)^2}$$

$$= \frac{1}{2x\sqrt{2\pi}} \int_{1/2}^2 e^{-\frac{1}{2}\left(\log \frac{x-1}{2}\right)^2} dx$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{1/2}^2 \frac{1}{x} e^{-\frac{1}{2}\left(t-\frac{1}{2}\right)^2} x dt$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-\ln 2}^{\ln 2} e^{-\frac{1}{2}\left(t-\frac{1}{2}\right)^2} dt$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-0.693}^{0.693} e^{-\frac{1}{2}\left(\frac{t-1}{2}\right)^2} dt$$

$$\frac{1}{2\sqrt{2\pi}} \int_{-0.8465}^{-0.1535} e^{-\frac{1}{2}z^2} dz$$

$$\frac{1}{\sqrt{2\pi}} \int_{-0.8465}^{-0.1535} e^{-\frac{1}{2}z^2} dz$$

$$= P(-0.8465 < z < -0.1535)$$

$$= \Phi(-0.1535) - (\Phi(-0.8465))$$

$$= 0.2996 - 0.0596$$

$$= 0.24 \quad \text{proved}$$

$$\log x = t$$

$$1/x = dt$$

$$dx = x dt$$

$$x = 1/2 \Rightarrow \log 1/2 = -\ln 2$$

$$x = 2 \Rightarrow t = \ln x$$

$$\frac{t-1}{2} = z$$

$$t-1 = 2z$$

$$t = 1 + 2z$$

$$dt = 2dz$$

$$t = -0.693$$

$$z = -0.8464$$

$$t = 0.693$$

$$z = -0.1535$$

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Q-NO. 04

Derive F-Distribution

Solution

$$f(x_1^2) = \frac{1}{2^{\frac{n_1}{2}} \sqrt{\frac{n_1}{2}}} e^{-\frac{x_1^2}{2}} (x_1^2)^{\frac{n_1}{2}-1} \quad 0 < x_1^2 < \infty$$

$$f(x_2^2) = \frac{1}{2^{\frac{n_2}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{x_2^2}{2}} (x_2^2)^{\frac{n_2}{2}-1} \quad 0 < x_2^2 < \infty$$

joint - Distribution

$$dF(x_1^2, x_2^2) = \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}, \frac{n_2}{2}}} e^{-\frac{1}{2}(x_1^2+x_2^2)} (x_1^2)^{\frac{n_1}{2}-1} (x_2^2)^{\frac{n_2}{2}-1} dx_1^2 dx_2^2$$

Transformation

$$F = \frac{x_1^2/n_1}{x_2^2/n_2} \quad \begin{array}{l} x_2^2 = v \\ dx_1^2 = \frac{n_2}{n_1} v F \end{array}$$

$$\frac{dx_1^2}{dF} = \frac{n_1}{n_2} v \quad \frac{dx_1^2}{dv} = \frac{n_1}{n_2} F$$

$$x_2^2 = v$$

$$\frac{dx_2^2}{dF} = 0$$

$$\frac{dx_2^2}{dv} = 1$$

Now

$$\left| \begin{array}{cc} \frac{n_1}{n_2} v & \frac{n_1}{n_2} F \\ 0 & 1 \end{array} \right|$$

$$J = \frac{n_1}{n_2} v$$

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Now

$$dF(x_1^2, x_2^2) = \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{1}{2}(x_1^2+x_2^2)} \left(\frac{n_1}{n_2} v F\right)^{\frac{n_1}{2}-1} (v)^{\frac{n_1}{2}-1} \frac{n_1}{n_2} v dv$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{1}{2}v(\frac{n_1}{n_2}F+1)} \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} \cdot v^{\frac{n_2}{2}+\frac{n_1}{2}-1} F^{\frac{n_1}{2}-1} dF dv$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1} dF \int_0^\infty v^{\frac{n_1+n_2}{2}} e^{-\frac{1}{2}v(1+\frac{n_1}{n_2}F)} dv$$

$$\frac{v}{2} (1 + \frac{n_1}{n_2} F) = z$$

$$v = \frac{2z}{1 + \frac{n_1}{n_2} F}$$

$$v=0 \quad z=0$$

$$v=\infty \quad z=\infty$$

$$dv = \frac{2}{1 + \frac{n_1}{n_2} F} dF$$

$$dP(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} dF \int_0^\infty e^{-z} \left(\frac{2z}{1 + \frac{n_1}{n_2} F}\right)^{\frac{n_1+n_2}{2}-1} \frac{2}{1 + \frac{n_1}{n_2} F} dz$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \left(\frac{2}{1 + \frac{n_1}{n_2} F}\right)^{\frac{n_1+n_2}{2}} dF \int_0^\infty e^{-z} z^{\frac{n_1+n_2}{2}-1} dz$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1} dF}{\sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}} (1 + \frac{n_1}{n_2} F)^{\frac{n_1+n_2}{2}}}$$

$$dpf = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right) (1 + \frac{n_1}{n_2} F)^{\frac{n_1+n_2}{2}}} dF$$

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$$= \frac{\binom{n_1}{\frac{n_1}{2}} \binom{n_2}{\frac{n_2}{2}} F^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right) (n_2+n_1 F)^{\frac{n_1+n_2}{2}}}$$

$$\frac{\binom{n_1}{\frac{n_1}{2}} \binom{n_2}{\frac{n_2}{2}} F^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right) (n_2+n_1 F)^{\frac{n_1+n_2}{2}}} \quad 0 < F < \infty$$

Characteristic function of
Cauchy Distribution

$$\phi(t) = E(e^{itx}) \quad f(x) = \frac{1}{\pi} \frac{\lambda}{\lambda^2 + (x-\mu)^2}$$

$$\phi(x) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

$$\int_{-\infty}^{\infty} e^{itx} \frac{1}{\pi} \frac{\lambda}{\lambda^2 + (x-\mu)^2} dx$$

$$\frac{1}{\pi} \int_{-\infty}^{\infty} e^{-txi} \frac{1}{\lambda^2 + (x-\mu)^2} dx$$

$$\int_{-\infty}^{\infty} e^{-itx} \frac{1}{\pi(1+x^2)} dx$$

$$\frac{1}{\pi} \int_{-\infty}^{\infty} e^{itx} \frac{1}{1+x^2} dx$$

By Laplace Distribution

$$\phi_2(t) = E(e^{itz})$$

$$= e^{it\mu} \phi(x)$$

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$$\int_{-\infty}^{\infty} e^{itz} \frac{1}{2} e^{-|z|} dz$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} e^{itz} e^{-|z|} dz$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\frac{1}{2} \int_{-\infty}^{\infty} (\cos tz + i \sin tz) \cdot e^{-|z|} dz$$

$$\frac{1}{2} \int_{-\infty}^{\infty} \cos tz e^{-|z|} dz + \frac{1}{2} \int_{-\infty}^{\infty} i \sin tz e^{-|z|} dz$$

$$= \int_{-\infty}^{\infty} \cos tz e^{-|z|} dz$$

$$\frac{m^2}{m^2 + 1}$$

$$\phi_2(t) = \frac{1}{1+t^2}$$

$$f(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itz} \phi_2(t) dt$$

$$\frac{1}{2} e^{-|z|} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itz} \frac{1}{1+t^2} dt$$

$$e^{-|z|} = \frac{1}{\pi} \int_{-\infty}^{\infty} e^{itz} \frac{1}{1+t^2} dt$$

$$e^{-|z|} = \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-tx} \frac{1}{1+x^2} dx$$

$$\phi_2(t) = e^{-|t|}$$

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IV

mean and variance

$$E(X) = m \quad v(X) = \sigma^2$$

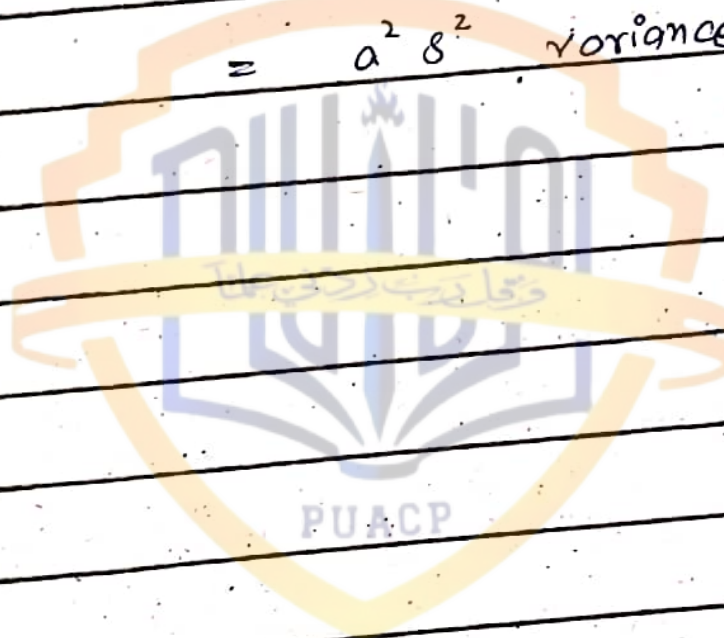
$$Y = aX + b$$

$$E(Y) = aE(X) + b$$

$$= am + b \quad \text{mean}$$

$$v(Y) = a^2 v(X)$$

$$= a^2 \sigma^2 \quad \text{variance}$$



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